

Exploring VPN Security
Techniques, Protocols, and
Attacks

Implementing VPN Solutions Project

**Secure Connectivity for
Modern Networks**





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VPN

A Virtual Private Network (VPN) is a secure connection between two or more devices over the internet, creating a private network

- **Purpose:**

- Privacy: Hides your IP address.
- Security: Encrypts data to prevent interception.
- Remote Access: Allows secure access to private networks.

- **Key Components:**

- Encryption: Secures data.
- Tunneling: Encapsulates data packets.
- Authentication: Verifies user identity.



VPN Types and Protocols

Remote Access VPN (Client-to-Site VPN)

Connects individual users securely to a company's internal network using VPN client software

Site-to-Site VPN

Connects two or more networks (e.g., branch offices) securely over the internet. Usually implemented between routers or firewalls

SSL VPN (Secure Sockets Layer VPN)

Uses HTTPS (port 443) to provide secure remote access via web browser or client. It encrypts traffic using SSL/TLS protocols.

IPsec VPN (Internet Protocol Security VPN)

Uses the IPsec protocol suite to encrypt and authenticate IP packets between sites or devices. It operates at the network layer

IMPLS VPN (Multiprotocol Label Switching VPN)

Uses service provider networks to create private paths for enterprise traffic, ensuring reliability and QoS.

L2TP/IPsec VPN

Combines Layer 2 Tunneling Protocol with IPsec encryption. Often used for secure remote access

SSL VPN

- **Overview**

- An SSL VPN allows secure remote access to a private network using a standard web browser or a lightweight VPN client. It operates over the SSL/TLS protocol, the same technology used to secure websites (HTTPS). Because it uses port 443, SSL VPN traffic can pass easily through most firewalls and network devices without requiring special configurations. SSL VPNs are widely used by organizations to enable remote employees, partners, or clients to securely access internal applications, files, or systems over the internet.

SSL VPN

- **Why SSL VPN?**

- Uses HTTPS (port 443) — easily passes through firewalls.
- Provides user-based authentication.
- Doesn't require a special client (can work in a browser).
- Offers tunnel mode (for full network access) and web mode (for portal-based access).

SSL VPN

- **How It Works**

- The user connects to the SSL VPN gateway (e.g., FortiGate) through a web browser or FortiClient VPN application.
- The connection is established using SSL/TLS encryption, ensuring data transmitted between the user and the VPN gateway is secure.
- Once authenticated, the user is granted access to internal resources depending on the assigned privileges.

Modes of SSL VPN

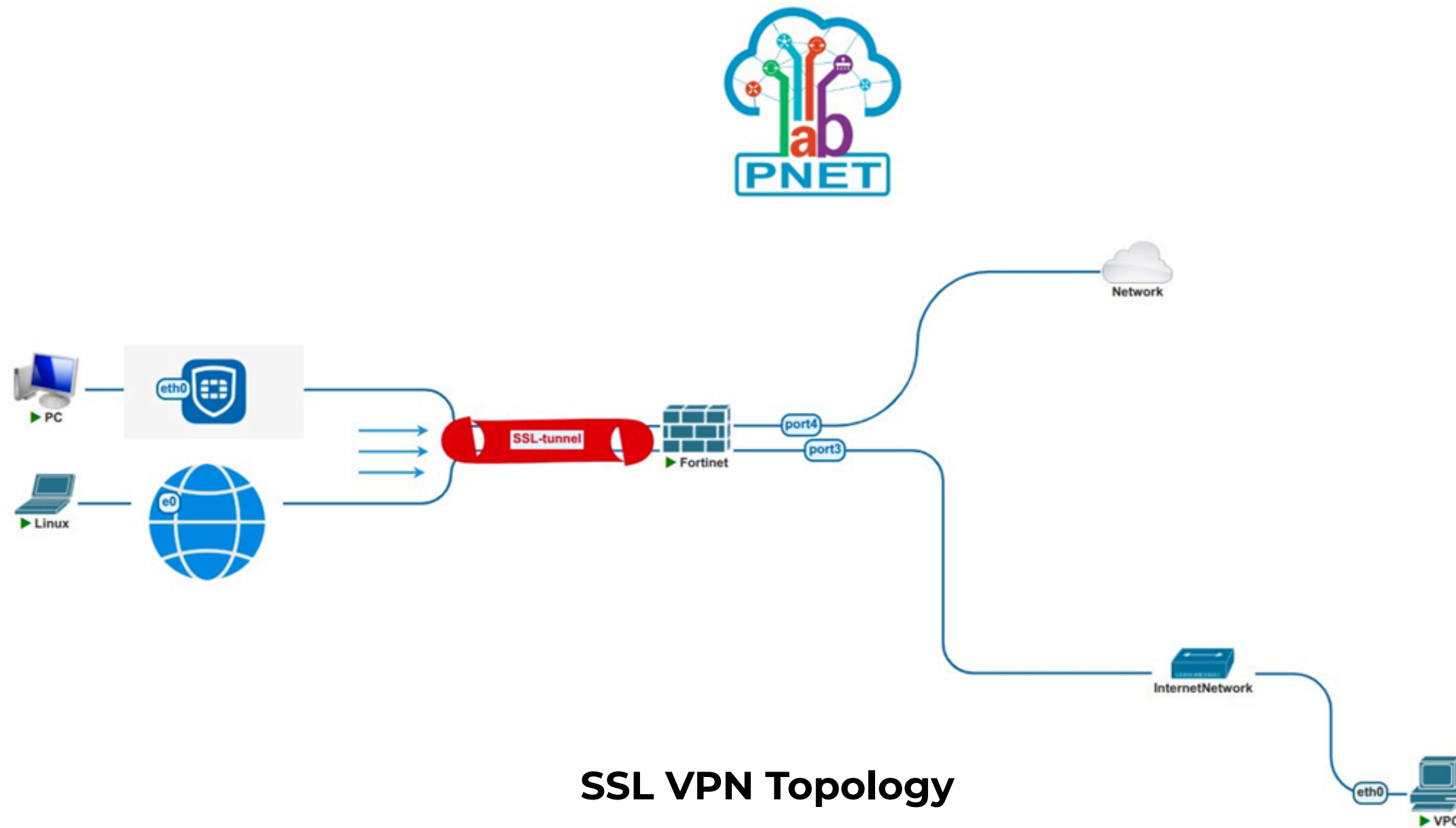
- **Web Mode**

- Access through a web portal.
- Users log in via a browser (e.g., <https://<public-ip>:443>) and can access specific applications like email, file servers, or web-based tools.
- Suitable for simple, controlled access without installing a VPN client

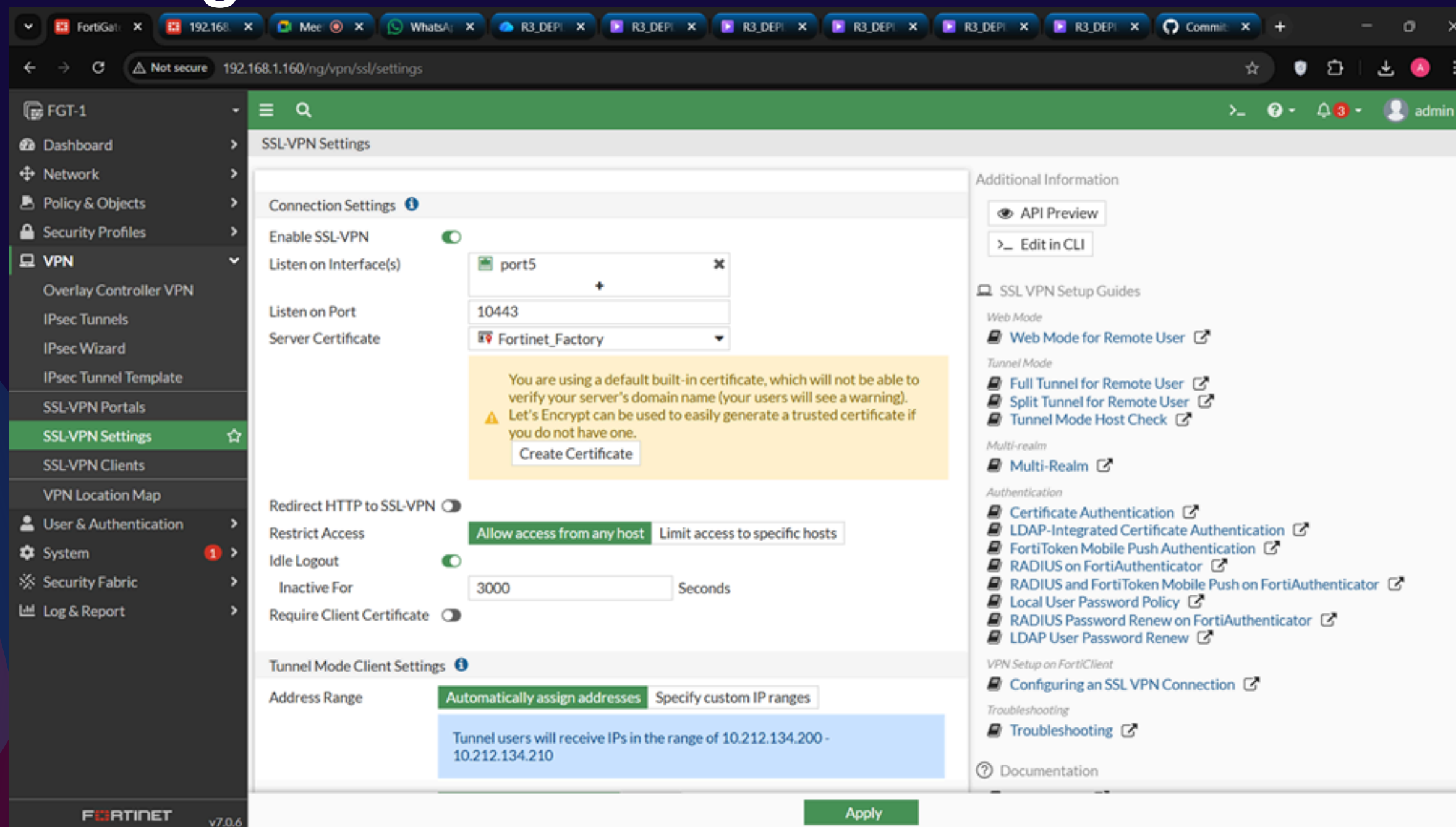
- **Tunnel Mode**

- Requires a VPN client (e.g., FortiClient).
- Creates a full network tunnel between the user's device and the internal network.
- Allows complete access to internal network resources as if the user were on-site.
- Ideal for IT staff or employees who need broader network access.

SSL VPN Configuration on FortiGate



SSL VPN Configuration on FortiGate



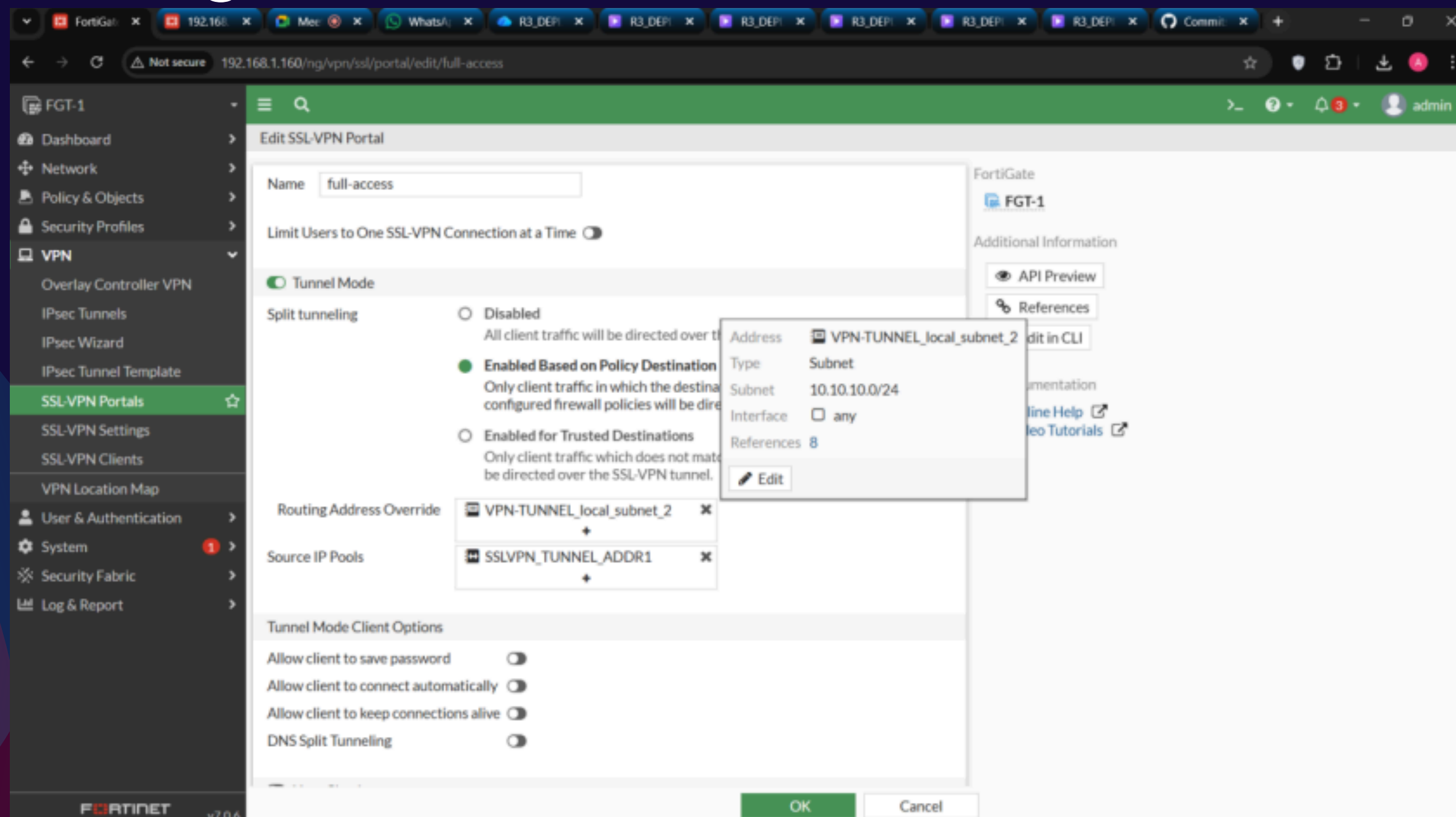
The screenshot displays the FortiGate web interface for configuring SSL-VPN settings. The left sidebar shows the navigation menu with 'VPN' expanded and 'SSL-VPN Settings' selected. The main content area is titled 'SSL-VPN Settings' and includes the following sections:

- Connection Settings:**
 - Enable SSL-VPN:** Toggled on.
 - Listen on Interface(s):** port5
 - Listen on Port:** 10443
 - Server Certificate:** Fortinet_Factory
 - Warning:** You are using a default built-in certificate, which will not be able to verify your server's domain name (your users will see a warning). Let's Encrypt can be used to easily generate a trusted certificate if you do not have one. [Create Certificate](#)
- Redirect HTTP to SSL-VPN:** Toggled on.
- Restrict Access:** [Allow access from any host](#) | [Limit access to specific hosts](#)
- Idle Logout:** Toggled on.
- Inactive For:** 3000 Seconds
- Require Client Certificate:** Toggled off.
- Tunnel Mode Client Settings:**
 - Address Range:** [Automatically assign addresses](#) | [Specify custom IP ranges](#)
 - Info:** Tunnel users will receive IPs in the range of 10.212.134.200 - 10.212.134.210

The right sidebar contains 'Additional Information' with links to 'API Preview' and 'Edit in CLI', and 'SSL VPN Setup Guides' categorized by Web Mode, Tunnel Mode, Multi-realm, Authentication, VPN Setup on FortiClient, and Troubleshooting.

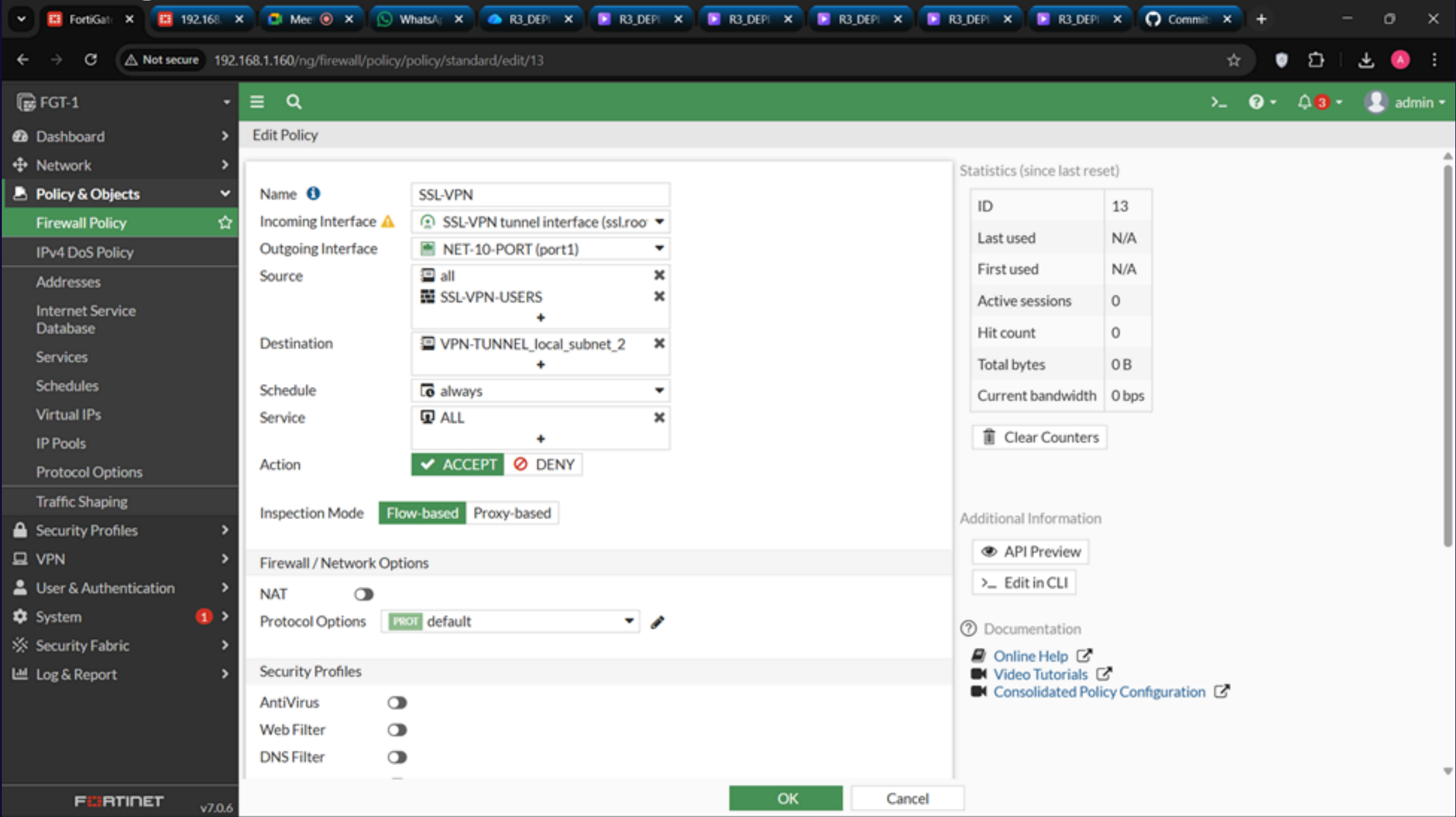
SSL-VPN Setting Configuration

SSL VPN Configuration on FortiGate



SSL-VPN Portal Configuration

SSL VPN Configuration on FortiGate



The screenshot displays the FortiGate Web UI configuration page for an SSL-VPN policy. The left sidebar shows the navigation menu with 'Policy & Objects' selected, and 'Firewall Policy' highlighted. The main content area is titled 'Edit Policy' and shows the configuration for a policy named 'SSL-VPN'. The configuration includes the following fields:

- Name:** SSL-VPN
- Incoming Interface:** SSL-VPN tunnel interface (ssl.root)
- Outgoing Interface:** NET-10-PORT (port1)
- Source:** all, SSL-VPN-USERS
- Destination:** VPN-TUNNEL_local_subnet_2
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT (checked), DENY
- Inspection Mode:** Flow-based (selected), Proxy-based
- Firewall / Network Options:**
 - NAT:** disabled
 - Protocol Options:** default
- Security Profiles:**
 - AntiVirus:** disabled
 - Web Filter:** disabled
 - DNS Filter:** disabled

On the right side, there is a 'Statistics (since last reset)' section with the following data:

ID	13
Last used	N/A
First used	N/A
Active sessions	0
Hit count	0
Total bytes	0 B
Current bandwidth	0 bps

Below the statistics, there is a 'Clear Counters' button. At the bottom right, there is an 'Additional Information' section with links for 'API Preview', 'Edit in CLI', 'Documentation', 'Online Help', 'Video Tutorials', and 'Consolidated Policy Configuration'. The bottom of the screen shows the 'OK' and 'Cancel' buttons.

SSL-VPN-Access Policy Configuration

IPsec VPN

- **Overview**

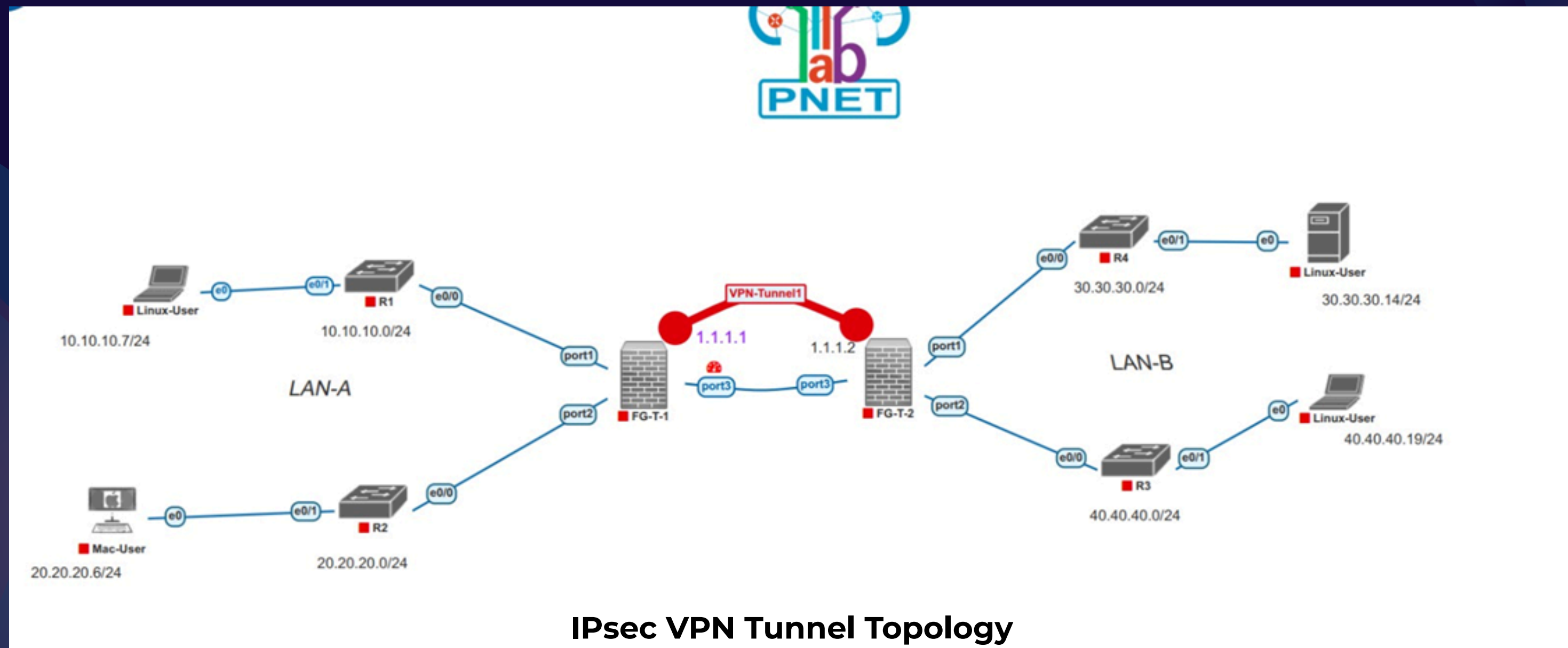
- The IPsec (Internet Protocol Security) VPN is a widely used protocol suite that secures network traffic by encrypting and authenticating IP packets at the network layer (Layer 3). It provides site-to-site and remote access solutions that ensure data confidentiality, integrity, and authenticity between two or more networks.
- IPsec VPNs are particularly suitable for organizations that need permanent, secure connections between branches, headquarters, or data centers.
- Unlike SSL VPNs, which are typically used for individual remote access, IPsec VPNs operate transparently for entire subnets and devices.

IPsec VPN

- **How It Works**

- An IPsec VPN establishes a secure tunnel between two FortiGate devices through a two-phase process:
 1. Phase 1 (IKE Negotiation):
 - Establishes a secure communication channel.
 - Authenticates the two VPN peers.
 - Negotiates encryption and hashing algorithms (e.g., AES, SHA).
 2. Phase 2 (Data Encryption):
 - Defines traffic selectors (networks that will communicate).
 - Encrypts user data packets through the established tunnel.

IPsec Configuration Steps on FortiGate



Testing and Verification

```
sakura
root@slitaz:~# ping 30.30.30.2
PING 30.30.30.2 (30.30.30.2): 56 data bytes
64 bytes from 30.30.30.2: seq=0 ttl=62 time=1.345 ms
64 bytes from 30.30.30.2: seq=1 ttl=62 time=1.014 ms
64 bytes from 30.30.30.2: seq=2 ttl=62 time=1.037 ms
64 bytes from 30.30.30.2: seq=3 ttl=62 time=1.154 ms
64 bytes from 30.30.30.2: seq=4 ttl=62 time=1.634 ms
64 bytes from 30.30.30.2: seq=5 ttl=62 time=0.821 ms
64 bytes from 30.30.30.2: seq=6 ttl=62 time=0.809 ms
64 bytes from 30.30.30.2: seq=7 ttl=62 time=0.661 ms
64 bytes from 30.30.30.2: seq=8 ttl=62 time=0.989 ms
64 bytes from 30.30.30.2: seq=9 ttl=62 time=0.895 ms

64 bytes from 30.30.30.2: seq=10 ttl=62 time=1.031 ms
64 bytes from 30.30.30.2: seq=11 ttl=62 time=2.264 ms
^C
--- 30.30.30.2 ping statistics ---
12 packets transmitted, 12 packets received, 0% packet loss
round-trip min/avg/max = 0.661/1.137/2.264 ms
root@slitaz:~#
```

Ping Test: ping from PC-1 to PC-3

Testing and Verification

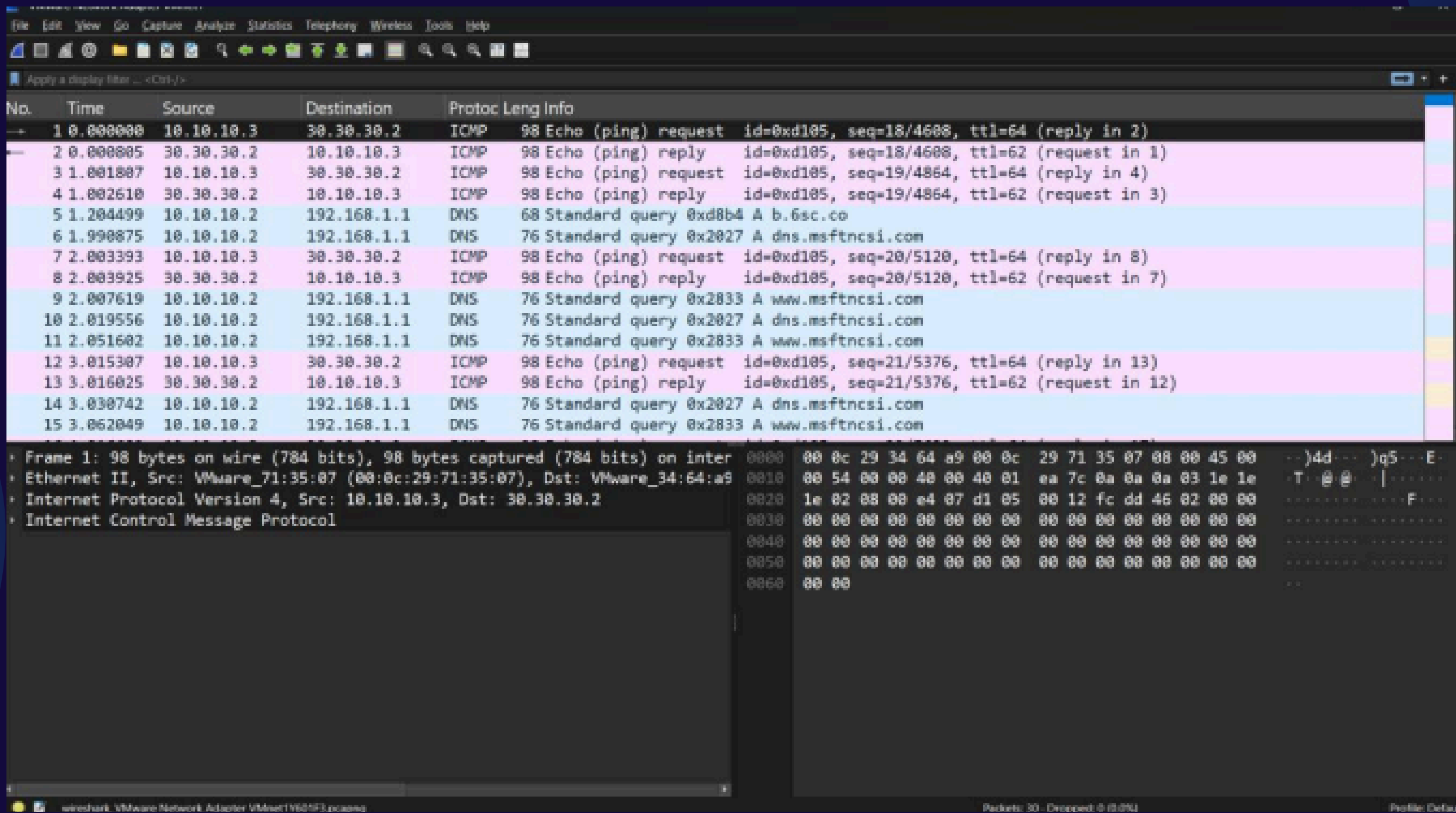
```
Thursday 13 November - 17:35

sakura

root@slitaz:~# ping 10.10.10.3
PING 10.10.10.3 (10.10.10.3): 56 data bytes
64 bytes from 10.10.10.3: seq=0 ttl=62 time=1.804 ms
64 bytes from 10.10.10.3: seq=1 ttl=62 time=1.001 ms
64 bytes from 10.10.10.3: seq=2 ttl=62 time=1.249 ms
64 bytes from 10.10.10.3: seq=3 ttl=62 time=1.493 ms
64 bytes from 10.10.10.3: seq=4 ttl=62 time=1.196 ms
64 bytes from 10.10.10.3: seq=5 ttl=62 time=1.080 ms
64 bytes from 10.10.10.3: seq=6 ttl=62 time=1.202 ms
64 bytes from 10.10.10.3: seq=7 ttl=62 time=1.440 ms
^C
--- 10.10.10.3 ping statistics ---
8 packets transmitted, 8 packets received, 0% packet loss
round-trip min/avg/max = 1.001/1.308/1.804 ms
root@slitaz:~#
```

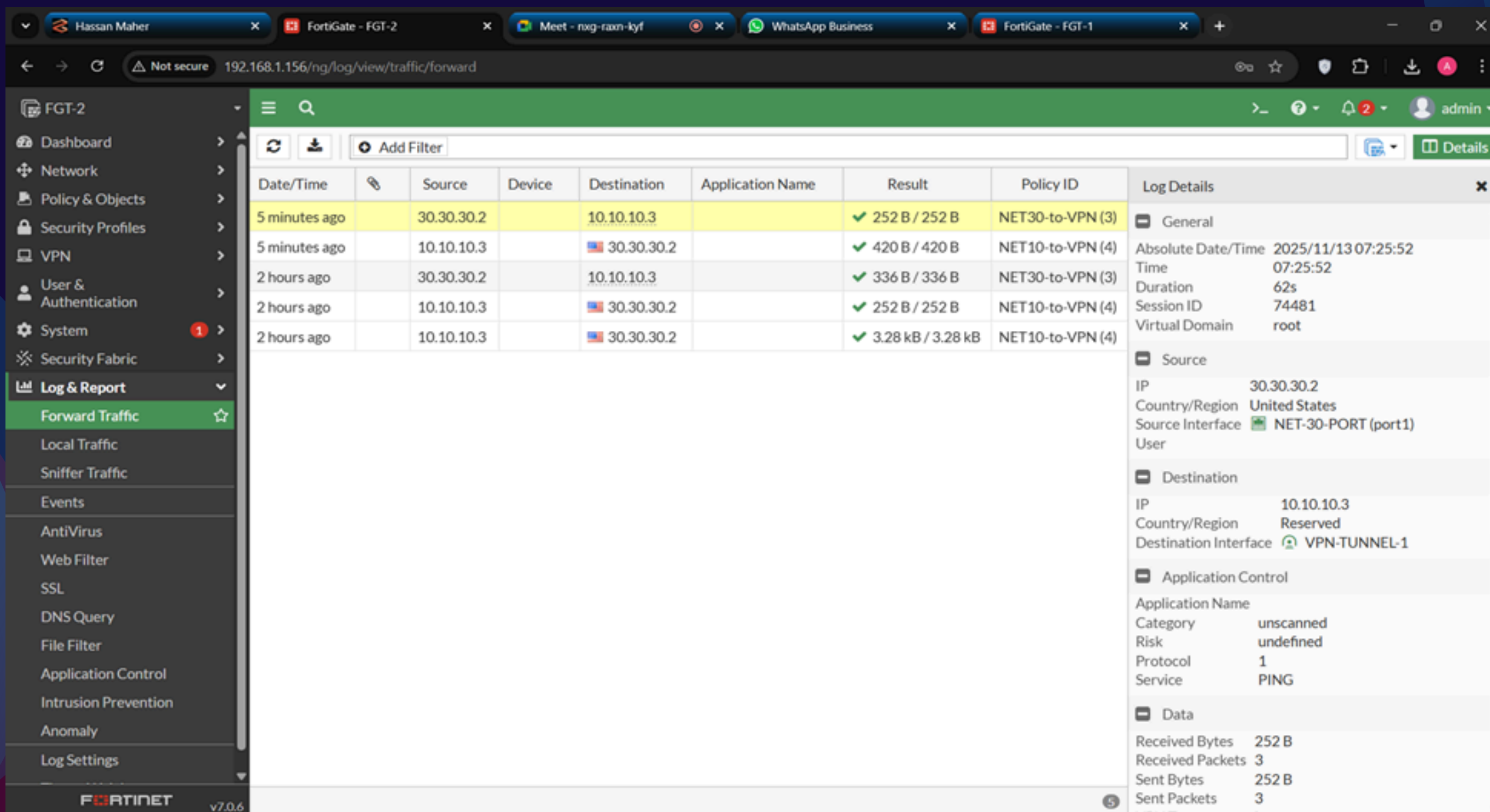
ping from PC-3 to PC-1

Testing and Verification



Capturing a traffic between PC-1 and PC-3 using Wireshark

Logs and Events



The screenshot displays the FortiGate web interface, specifically the 'Log & Report' section. The left sidebar shows the navigation menu with 'Log & Report' selected. The main content area shows a table of traffic logs with columns: Date/Time, Source, Device, Destination, Application Name, Result, and Policy ID. The table lists several log entries, with the first one highlighted. To the right of the table, a detailed view of the selected log entry is shown, including General information (Absolute Date/Time, Time, Duration, Session ID, Virtual Domain), Source information (IP, Country/Region, Source Interface, User), Destination information (IP, Country/Region, Destination Interface), Application Control information (Application Name, Category, Risk, Protocol, Service), and Data information (Received Bytes, Received Packets, Sent Bytes, Sent Packets).

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID
5 minutes ago	30.30.30.2		10.10.10.3		✓ 252 B / 252 B	NET30-to-VPN (3)
5 minutes ago	10.10.10.3		30.30.30.2		✓ 420 B / 420 B	NET10-to-VPN (4)
2 hours ago	30.30.30.2		10.10.10.3		✓ 336 B / 336 B	NET30-to-VPN (3)
2 hours ago	10.10.10.3		30.30.30.2		✓ 252 B / 252 B	NET10-to-VPN (4)
2 hours ago	10.10.10.3		30.30.30.2		✓ 3.28 kB / 3.28 kB	NET10-to-VPN (4)

Log Details

General

- Absolute Date/Time: 2025/11/13 07:25:52
- Time: 07:25:52
- Duration: 62s
- Session ID: 74481
- Virtual Domain: root

Source

- IP: 30.30.30.2
- Country/Region: United States
- Source Interface: NET-30-PORT (port1)
- User:

Destination

- IP: 10.10.10.3
- Country/Region: Reserved
- Destination Interface: VPN-TUNNEL-1

Application Control

- Application Name:
- Category: unscanned
- Risk: undefined
- Protocol: 1
- Service: PING

Data

- Received Bytes: 252 B
- Received Packets: 3
- Sent Bytes: 252 B
- Sent Packets: 3

Tunnel Status

FortiGate - FGT-1 x FortiGate - FGT-2 x Google Meet x LAB 10: SSL VPN x OSB download | SourceForge.n x

Not secure 192.168.1.159/ng/vpn/ipsec

admin

FortiGate time is out of sync.

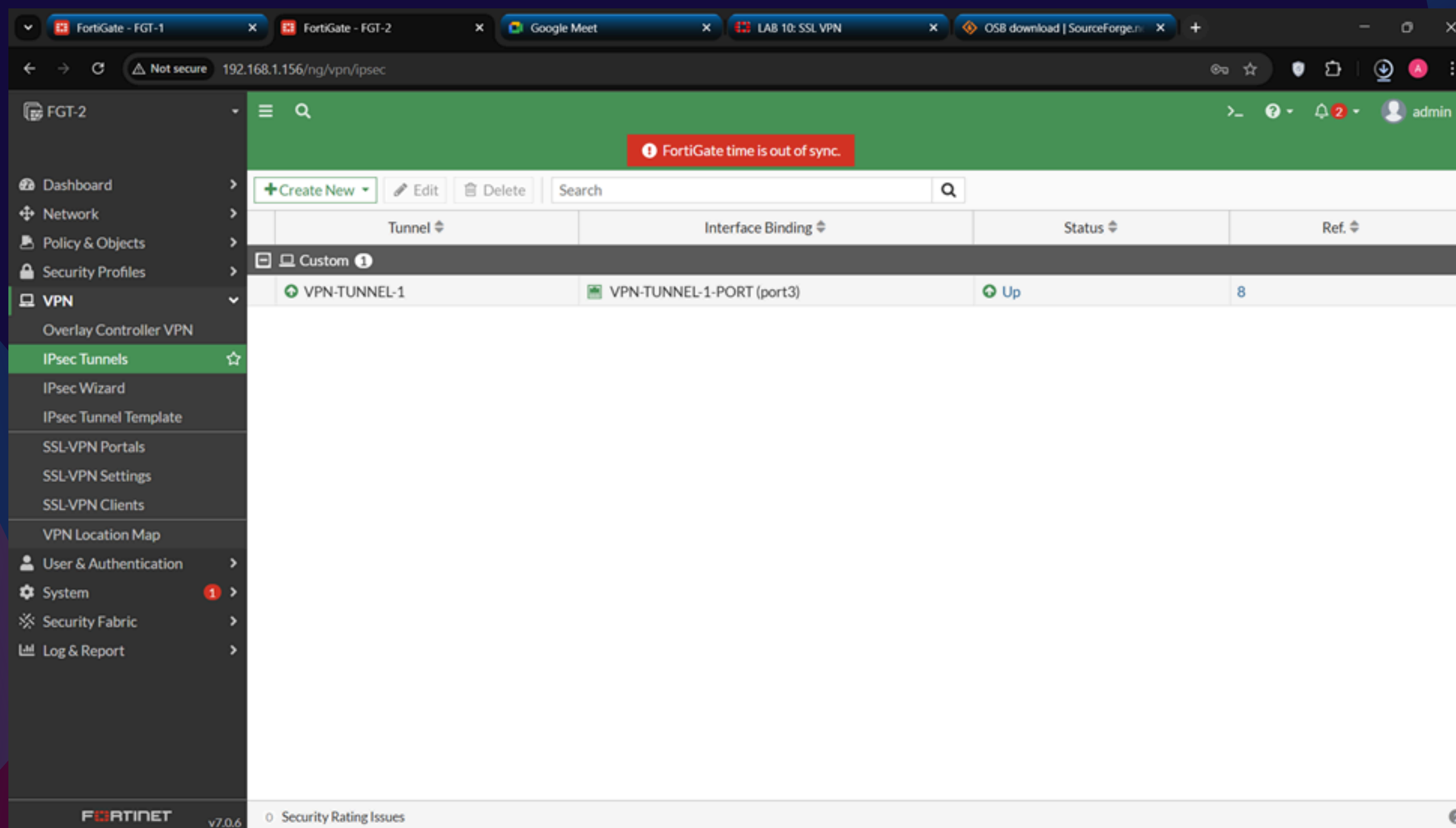
+ Create New Edit Delete Search

Tunnel	Interface Binding	Status	Ref.
VPN-TUNNEL-1	VPN-TUNNEL-1-PORT (port3)	Up	8

Dashboard Network Policy & Objects Security Profiles VPN Overlay Controller VPN IPsec Tunnels IPsec Wizard IPsec Tunnel Template SSL-VPN Portals SSL-VPN Settings SSL-VPN Clients VPN Location Map User & Authentication System Security Fabric Log & Report

FORTINET v7.0.6 0 Security Rating Issues

Tunnel Status



FortiGate - FGT-1 FortiGate - FGT-2 Google Meet LAB 10: SSL VPN OSB download | SourceForge.net

Not secure 192.168.1.156/ng/vpn/ipsec

FGT-2

Dashboard Network Policy & Objects Security Profiles VPN Overlay Controller VPN IPsec Tunnels IPsec Wizard IPsec Tunnel Template SSL-VPN Portals SSL-VPN Settings SSL-VPN Clients VPN Location Map User & Authentication System Security Fabric Log & Report

FortiGate time is out of sync.

Create New Edit Delete Search

Tunnel	Interface Binding	Status	Ref.
VPN-TUNNEL-1	VPN-TUNNEL-1-PORT (port3)	Up	8

FORTINET v7.0.6 0 Security Rating Issues

FortiGate VPN with SD-WAN Integration

- **Overview**

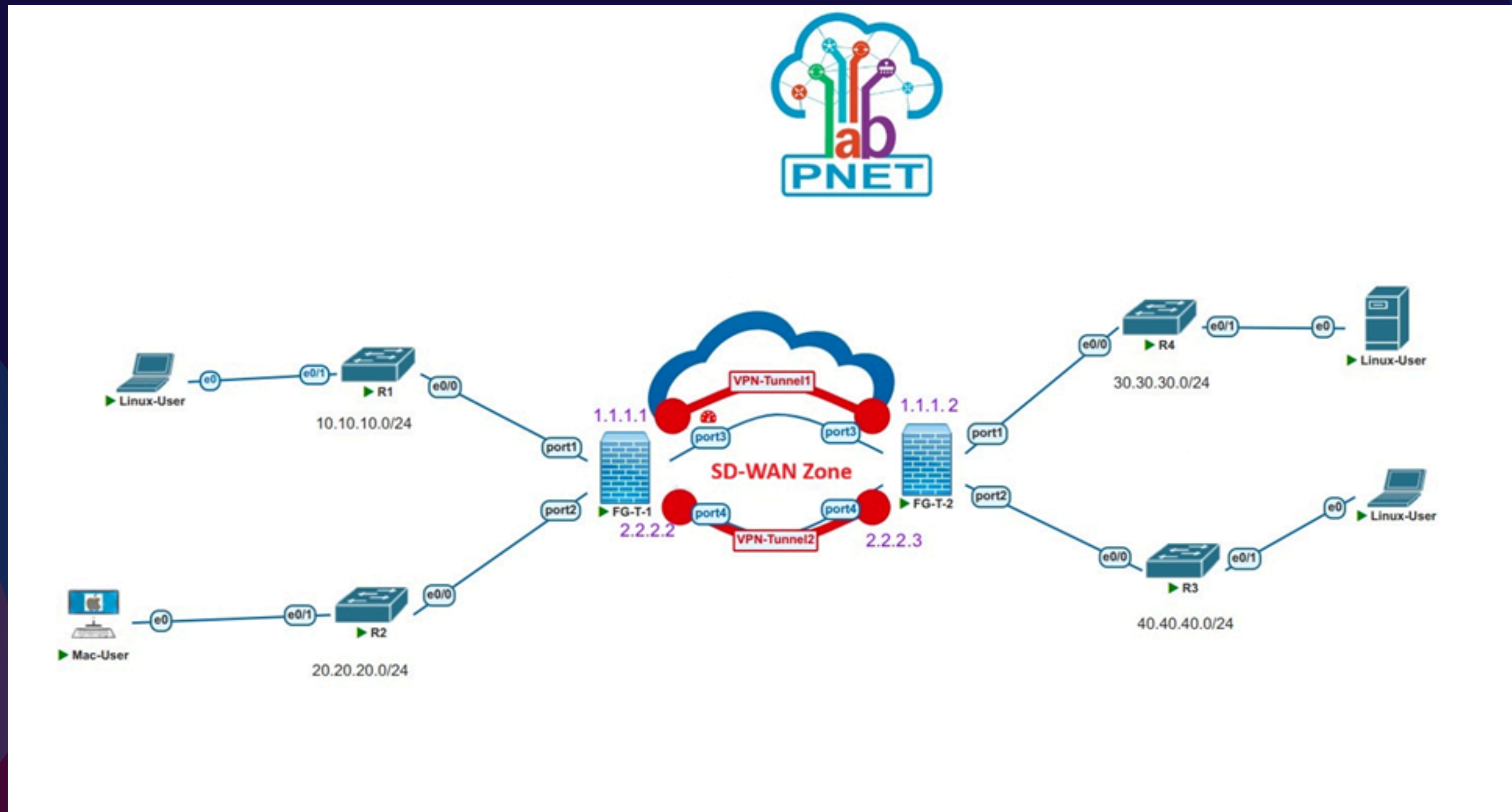
SD-WAN (Software-Defined Wide Area Network) is an advanced networking technology that enhances WAN performance by intelligently directing traffic across multiple WAN connections (such as MPLS, broadband, or LTE). Integrating SD-WAN with VPN allows FortiGate devices to dynamically select the best path for VPN traffic based on real-time performance metrics.

FortiGate VPN with SD-WAN Integration

- **How SD-WAN Works**

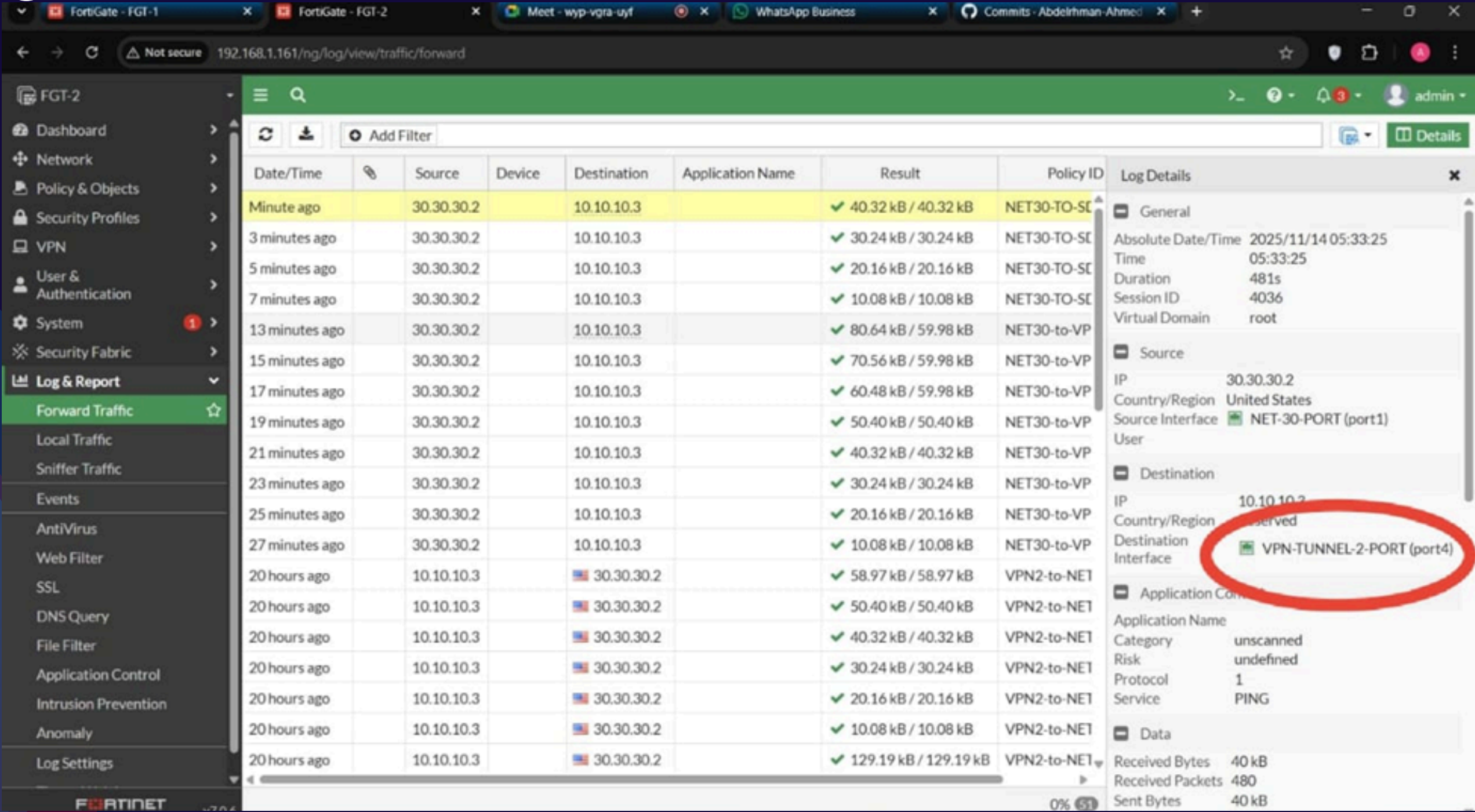
SD-WAN operates by creating a virtual overlay network on top of existing physical connections like broadband, MPLS, or LTE. Edge devices installed at branch offices, data centers, or cloud locations manage and route traffic according to centrally defined policies.

SD-WAN Configuration on FortiGate



VPN with SD-WAN Integration Topology

Testing and Performance Evaluation

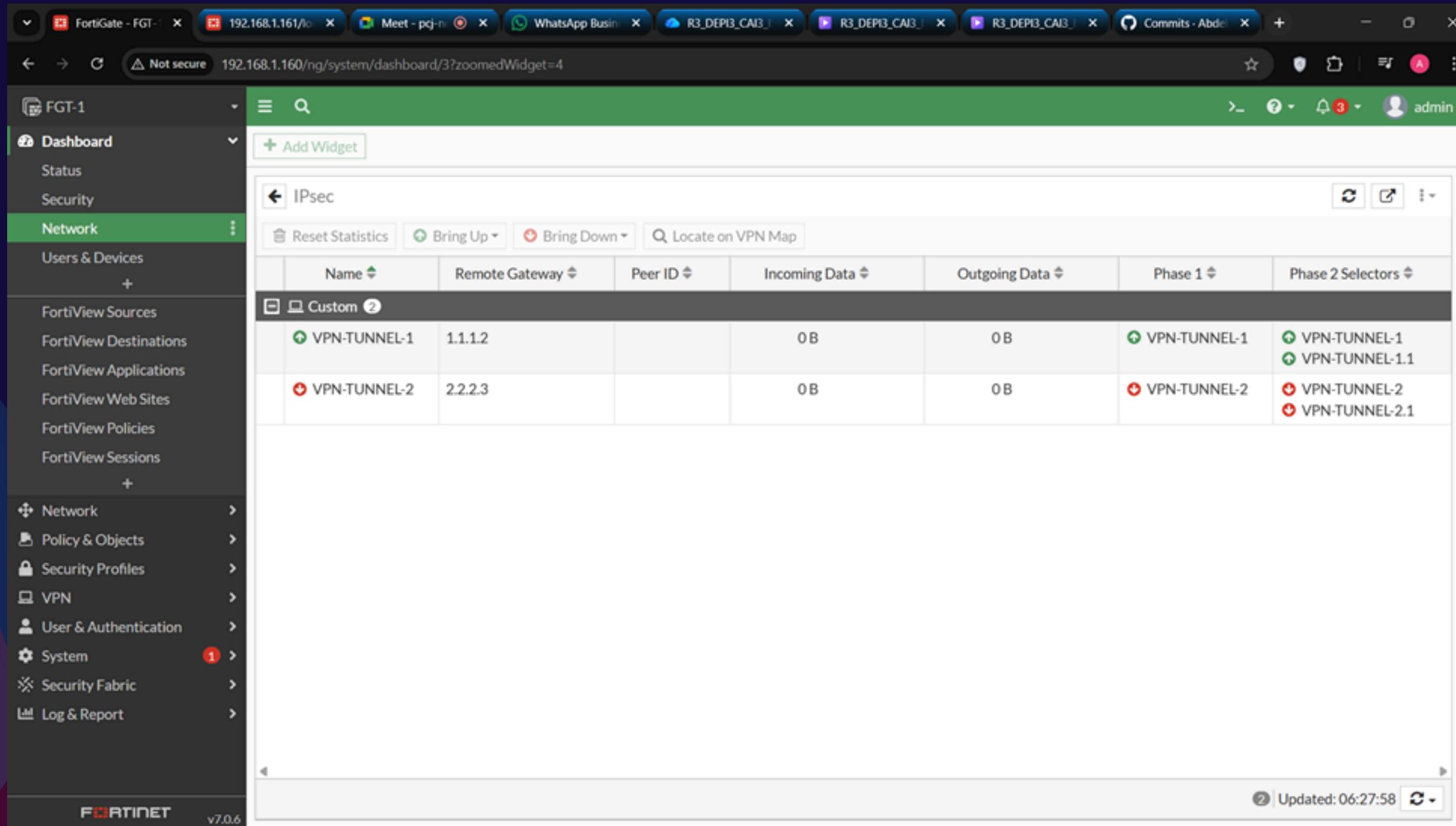


The screenshot displays the Fortinet FortiGate web interface. The left sidebar shows the navigation menu with 'Log & Report' selected. The main area shows a table of traffic logs. The first log entry is highlighted in yellow. The right sidebar shows the details for the selected log entry, with 'VPN-TUNNEL-2-PORT (port4)' circled in red.

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID	Log Details
Minute ago	30.30.30.2		10.10.10.3		✓ 40.32 kB / 40.32 kB	NET30-TO-SC	General Absolute Date/Time 2025/11/14 05:33:25 Time 05:33:25 Duration 481s Session ID 4036 Virtual Domain root
3 minutes ago	30.30.30.2		10.10.10.3		✓ 30.24 kB / 30.24 kB	NET30-TO-SC	Source IP 30.30.30.2 Country/Region United States Source Interface NET-30-PORT (port1) User
5 minutes ago	30.30.30.2		10.10.10.3		✓ 20.16 kB / 20.16 kB	NET30-TO-SC	Destination IP 10.10.10.3 Country/Region reserved Destination Interface VPN-TUNNEL-2-PORT (port4)
7 minutes ago	30.30.30.2		10.10.10.3		✓ 10.08 kB / 10.08 kB	NET30-TO-SC	Application Control Application Name Category unscanned Risk undefined Protocol 1 Service PING
13 minutes ago	30.30.30.2		10.10.10.3		✓ 80.64 kB / 59.98 kB	NET30-to-VP	Data Received Bytes 40 kB Received Packets 480 Sent Bytes 40 kB
15 minutes ago	30.30.30.2		10.10.10.3		✓ 70.56 kB / 59.98 kB	NET30-to-VP	
17 minutes ago	30.30.30.2		10.10.10.3		✓ 60.48 kB / 59.98 kB	NET30-to-VP	
19 minutes ago	30.30.30.2		10.10.10.3		✓ 50.40 kB / 50.40 kB	NET30-to-VP	
21 minutes ago	30.30.30.2		10.10.10.3		✓ 40.32 kB / 40.32 kB	NET30-to-VP	
23 minutes ago	30.30.30.2		10.10.10.3		✓ 30.24 kB / 30.24 kB	NET30-to-VP	
25 minutes ago	30.30.30.2		10.10.10.3		✓ 20.16 kB / 20.16 kB	NET30-to-VP	
27 minutes ago	30.30.30.2		10.10.10.3		✓ 10.08 kB / 10.08 kB	NET30-to-VP	
20 hours ago	10.10.10.3		30.30.30.2		✓ 58.97 kB / 58.97 kB	VPN2-to-NET	
20 hours ago	10.10.10.3		30.30.30.2		✓ 50.40 kB / 50.40 kB	VPN2-to-NET	
20 hours ago	10.10.10.3		30.30.30.2		✓ 40.32 kB / 40.32 kB	VPN2-to-NET	
20 hours ago	10.10.10.3		30.30.30.2		✓ 30.24 kB / 30.24 kB	VPN2-to-NET	
20 hours ago	10.10.10.3		30.30.30.2		✓ 20.16 kB / 20.16 kB	VPN2-to-NET	
20 hours ago	10.10.10.3		30.30.30.2		✓ 10.08 kB / 10.08 kB	VPN2-to-NET	
20 hours ago	10.10.10.3		30.30.30.2		✓ 129.19 kB / 129.19 kB	VPN2-to-NET	

1. Tunnel 1 down and Tunnel 2 up

Testing and Performance Evaluation



The screenshot shows the FortiGate web interface with the 'Network' tab selected. The 'IPsec' section is active, displaying a table of VPN tunnels. Tunnel 1 is up and operational, while Tunnel 2 is down. The table includes columns for Name, Remote Gateway, Peer ID, Incoming Data, Outgoing Data, Phase 1, and Phase 2 Selectors.

Name	Remote Gateway	Peer ID	Incoming Data	Outgoing Data	Phase 1	Phase 2 Selectors
VPN-TUNNEL-1	1.1.1.2		0B	0B	VPN-TUNNEL-1	VPN-TUNNEL-1 VPN-TUNNEL-1.1
VPN-TUNNEL-2	2.2.2.3		0B	0B	VPN-TUNNEL-2	VPN-TUNNEL-2 VPN-TUNNEL-2.1

2. Tunnel 1 up and Tunnel 2 down

Testing and Performance Evaluation

FGT-1

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

SD-WAN Zones

SD-WAN Rules

Performance SLAs

Packet Loss

Latency

Jitter

No data

Create New

Edit

Delete

Search

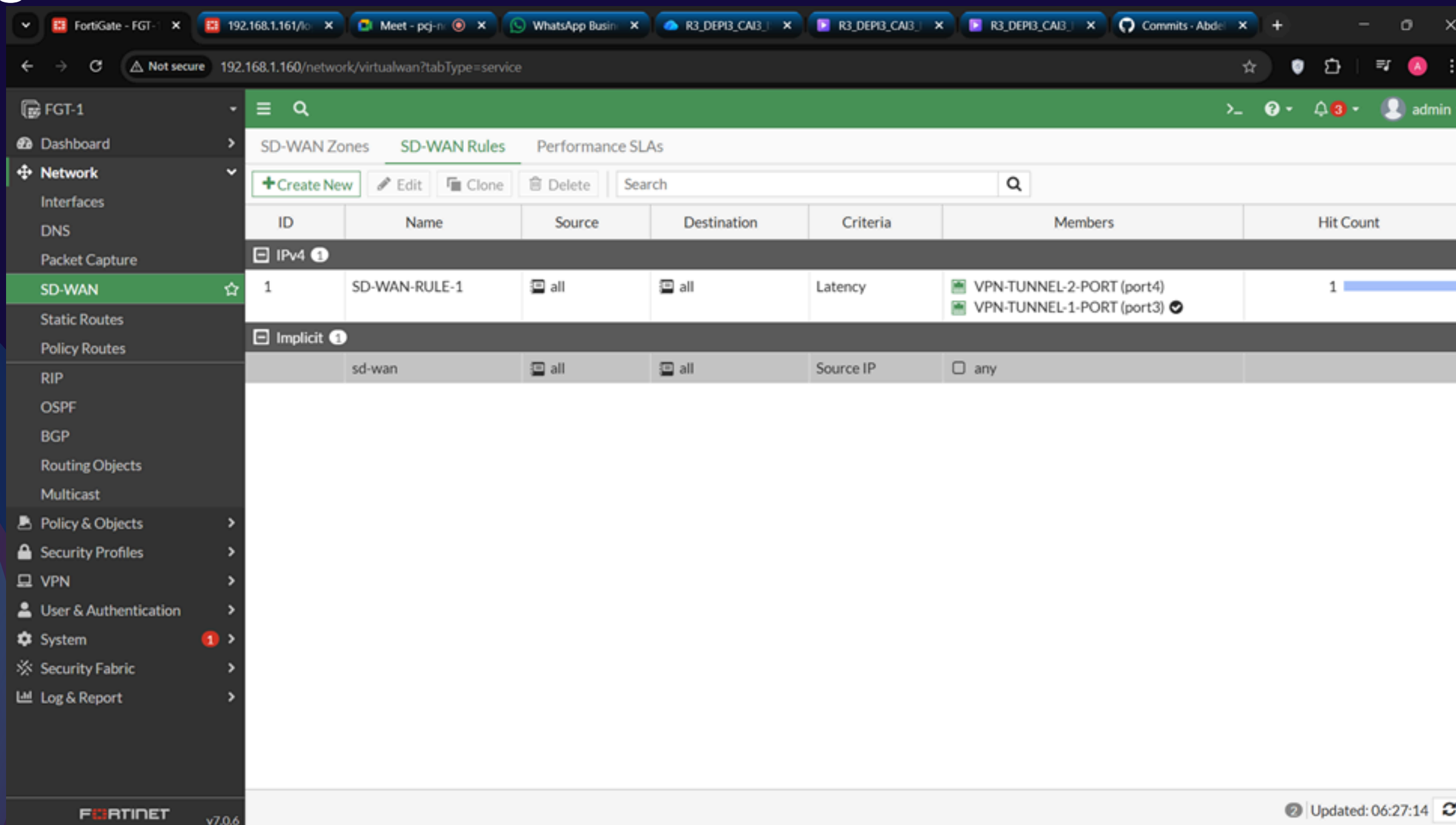
Name	Detect Server	Packet Loss	Latency	Jitter	Failure Thres
Default_AWS	http://aws.amazon.com/				5
Default_DNS	96.45.45.45 96.45.46.46 (System DNS)				5
Default_FortiGuard	http://fortiguard.com/				5
Default_Gmail	gmail.com				5
Default_Google Search	http://www.google.com/				5
Default_Office_365	http://www.office.com/				5
SLA_PROFILE	1.1.1.2 2.2.2.3	VPN-TUNNEL-1-PORT (port3): 0.00% VPN-TUNNEL-2-PORT (port4):	VPN-TUNNEL-1-PORT (port3): 0.33ms VPN-TUNNEL-2-PORT (port4):	VPN-TUNNEL-1-PORT (port3): 0.11ms VPN-TUNNEL-2-PORT (port4):	5

FORTINET

v7.0.6

2. Tunnel 1 up and Tunnel 2 down

Testing and Performance Evaluation



The screenshot shows the FortiGate SD-WAN Rules configuration page. The left sidebar contains the navigation menu with 'SD-WAN' selected. The main area displays a table of SD-WAN Rules. The table has columns for ID, Name, Source, Destination, Criteria, Members, and Hit Count. There are two rules listed: 'SD-WAN-RULE-1' and 'Implicit'. The 'SD-WAN-RULE-1' rule is selected and shows its configuration details in the right pane.

ID	Name	Source	Destination	Criteria	Members	Hit Count
1	SD-WAN-RULE-1	all	all	Latency	VPN-TUNNEL-2-PORT (port4) VPN-TUNNEL-1-PORT (port3)	1
	Implicit	sd-wan	all	Source IP	any	

2. Tunnel 1 up and Tunnel 2 down

Testing and Performance Evaluation

FGT-1

Dashboard

Network

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

Forward Traffic

Local Traffic

Sniffer Traffic

Events

AntiVirus

Web Filter

SSL

DNS Query

File Filter

Application Control

Intrusion Prevention

Anomaly

Log Settings

192.168.1.160/hg/log/view/traffic/forward

admin

Add Filter

Date/Time	Source	Device	Destination	Application Name	Result	Policy ID
Minute ago	30.30.30.2		10.10.10.3		✓ 30.24 kB / 30.24 kB	SD-WAN-TO
3 minutes ago	30.30.30.2		10.10.10.3		✓ 20.16 kB / 20.16 kB	SD-WAN-TO
5 minutes ago	30.30.30.2		10.10.10.3		✓ 10.08 kB / 10.08 kB	SD-WAN-TO
9 minutes ago	30.30.30.2		10.10.10.3		✓ 110.88 kB / 110.88 kB	SD-WAN-TO
11 minutes ago	30.30.30.2		10.10.10.3		✓ 100.80 kB / 100.80 kB	SD-WAN-TO
13 minutes ago	30.30.30.2		10.10.10.3		✓ 90.72 kB / 90.72 kB	SD-WAN-TO
15 minutes ago	30.30.30.2		10.10.10.3		✓ 80.64 kB / 80.64 kB	SD-WAN-TO
17 minutes ago	30.30.30.2		10.10.10.3		✓ 70.56 kB / 70.56 kB	SD-WAN-TO
19 minutes ago	30.30.30.2		10.10.10.3		✓ 60.48 kB / 60.48 kB	SD-WAN-TO
21 minutes ago	30.30.30.2		10.10.10.3		✓ 50.40 kB / 50.40 kB	SD-WAN-TO
23 minutes ago	30.30.30.2		10.10.10.3		✓ 40.32 kB / 40.32 kB	SD-WAN-TO
25 minutes ago	30.30.30.2		10.10.10.3		✓ 30.24 kB / 30.24 kB	SD-WAN-TO
27 minutes ago	30.30.30.2		10.10.10.3		✓ 20.16 kB / 20.16 kB	SD-WAN-TO
29 minutes ago	30.30.30.2		10.10.10.3		✓ 10.08 kB / 10.08 kB	SD-WAN-TO
32 minutes ago	30.30.30.2		10.10.10.3		✓ 141.12 kB / 141.12 kB	SD-WAN-TO
34 minutes ago	30.30.30.2		10.10.10.3		✓ 131.04 kB / 131.04 kB	SD-WAN-TO
36 minutes ago	30.30.30.2		10.10.10.3		✓ 120.96 kB / 120.96 kB	SD-WAN-TO
38 minutes ago	30.30.30.2		10.10.10.3		✓ 110.88 kB / 110.88 kB	SD-WAN-TO
40 minutes ago	30.30.30.2		10.10.10.3		✓ 100.80 kB / 100.80 kB	SD-WAN-TO

Log Details

General

Absolute Date/Time 2025/11/14 06:25:03

Time 06:25:03

Duration 361s

Session ID 80

Virtual Domain root

Source

IP 30.30.30.2

Country/Region Reserved

Source Interface VPN-TUNNEL-1-PORT (port3)

User

Destination

IP 10.10.10.3

Country/Region Reserved

Destination Interface NET-10-PORT (port1)

Application Control

Application Name

Category unscanned

Risk undefined

Protocol 1

Service PING

Data

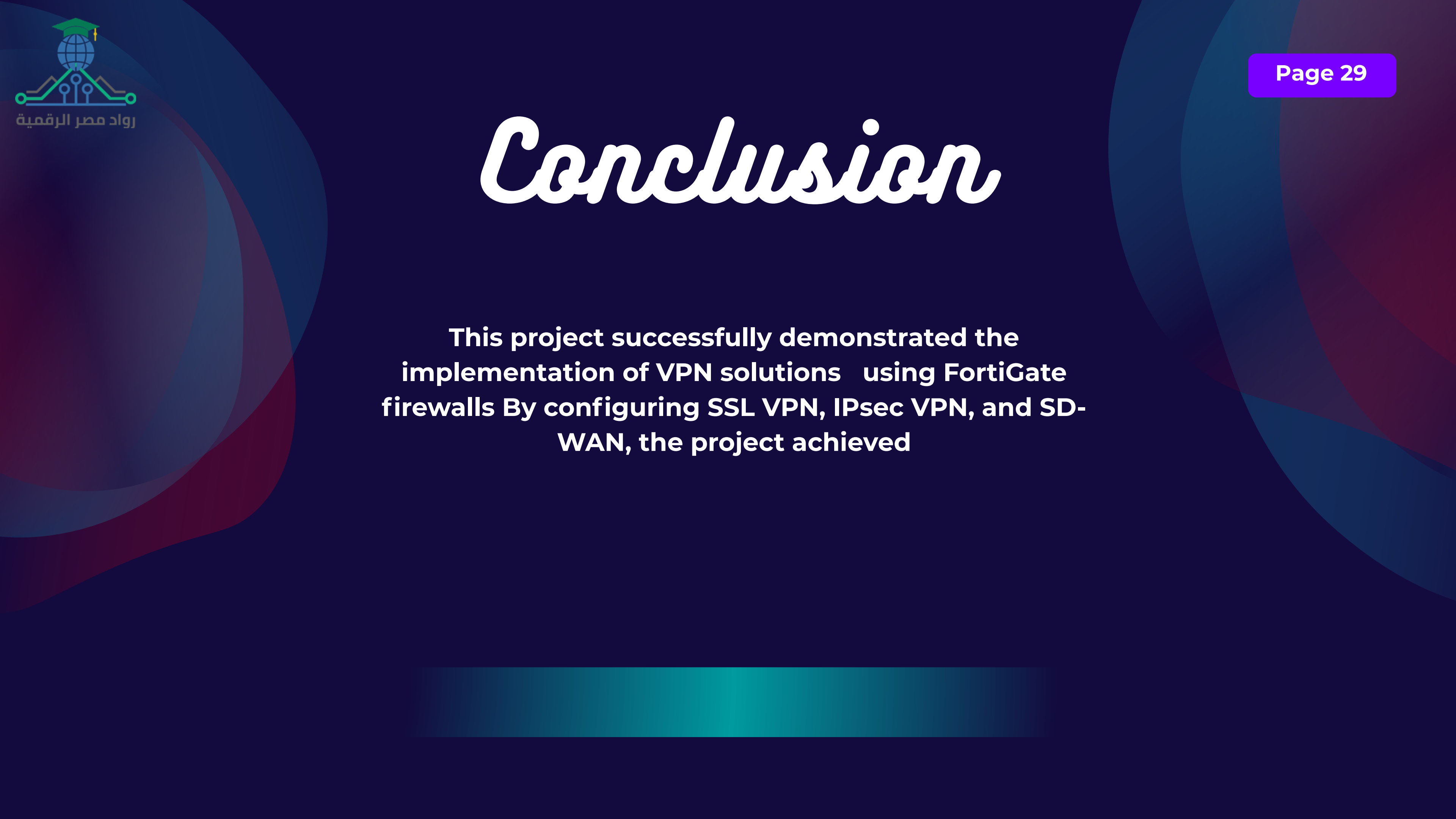
Received Bytes 30 kB

Received Packets 360

Sent Bytes 30 kB

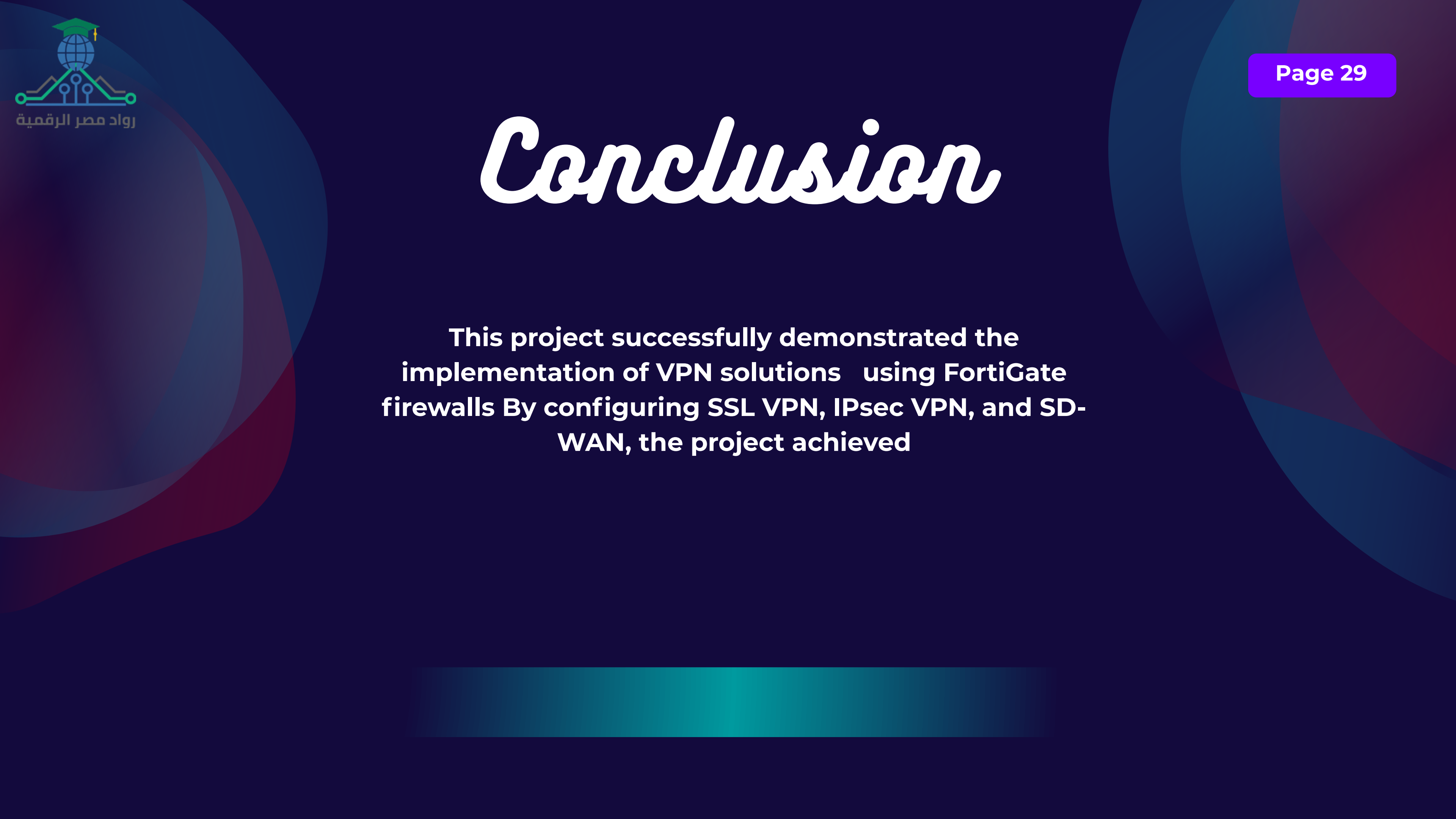
Sent Packets 360

2. Tunnel 1 up and Tunnel 2 down



Conclusion

This project successfully demonstrated the implementation of VPN solutions using FortiGate firewalls By configuring SSL VPN, IPsec VPN, and SD-WAN, the project achieved



Thank you